

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Full Stack Development (FSD) Documentation

ShopEZ — MERN Stack E-commerce Application

Phase: Project Documentation

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Classification: Project Documentation — For Academic Submission

Full Stack Development with MERN — Project Documentation

1. Executive Summary

ShopEZ is a production-grade, full-stack e-commerce application developed using the MERN (MongoDB, Express.js, React.js, Node.js) technology stack. The platform provides a comprehensive retail solution encompassing customer-facing product discovery, cart management, and payment processing, alongside a dedicated administrative portal for inventory, order fulfilment, and business analytics. Deployed on Vercel (frontend) and Render (backend) with MongoDB Atlas as the cloud database.

2. System Architecture

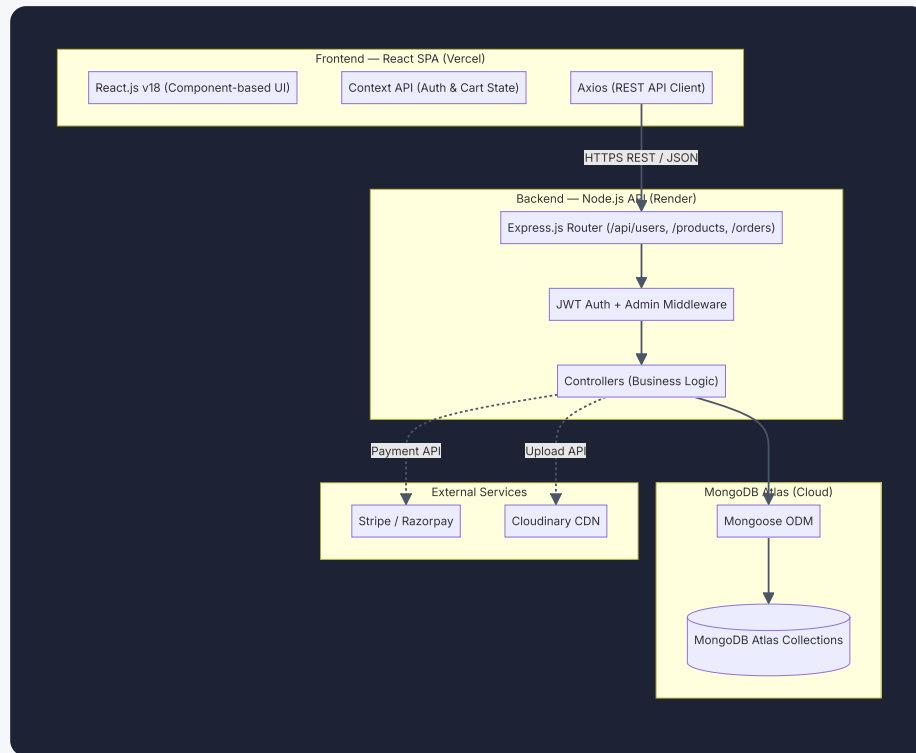


FIGURE 11.1 — FULL SYSTEM ARCHITECTURE

3. Database ER Diagram

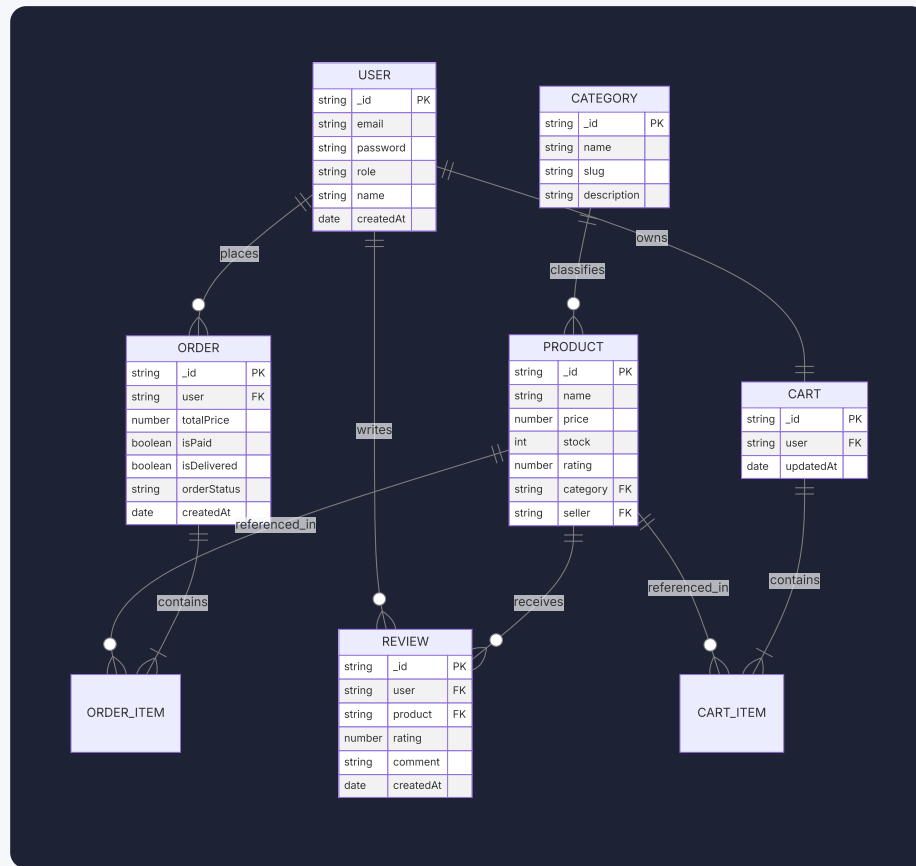


FIGURE 11.2 — ENTITY RELATIONSHIP DIAGRAM

4. UML Class Diagram — Mongoose Models

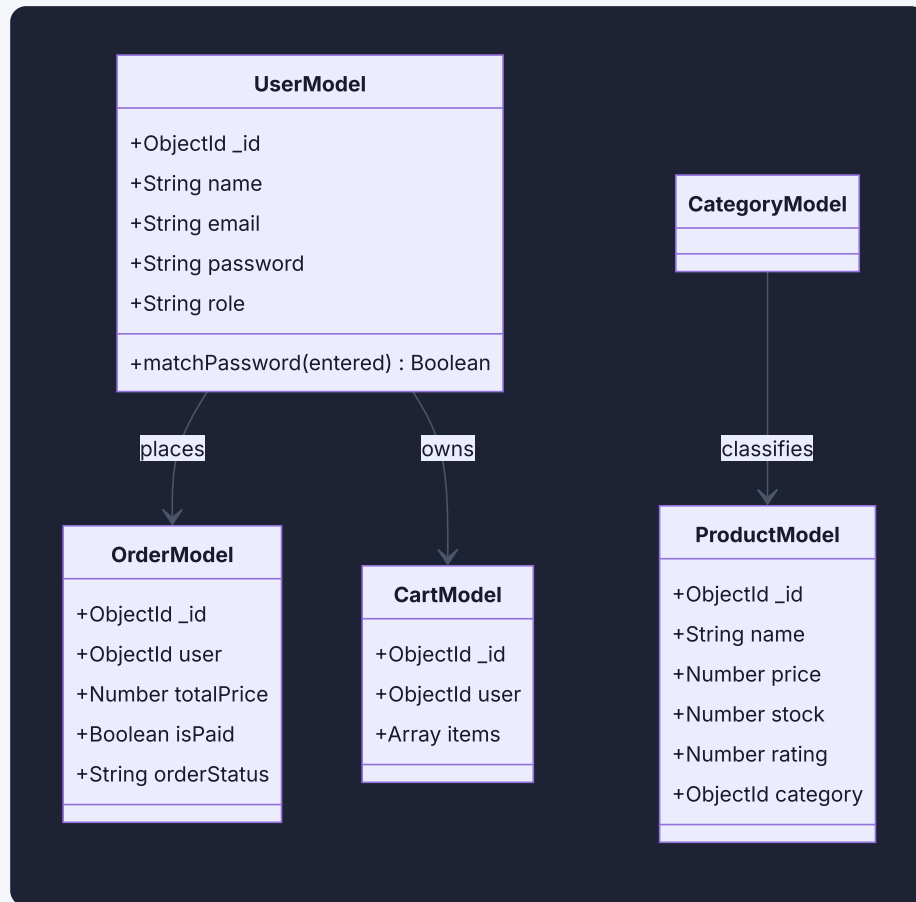


FIGURE 11.3 — UML CLASS DIAGRAM: CORE DATA MODELS

5. Authentication Sequence Diagram

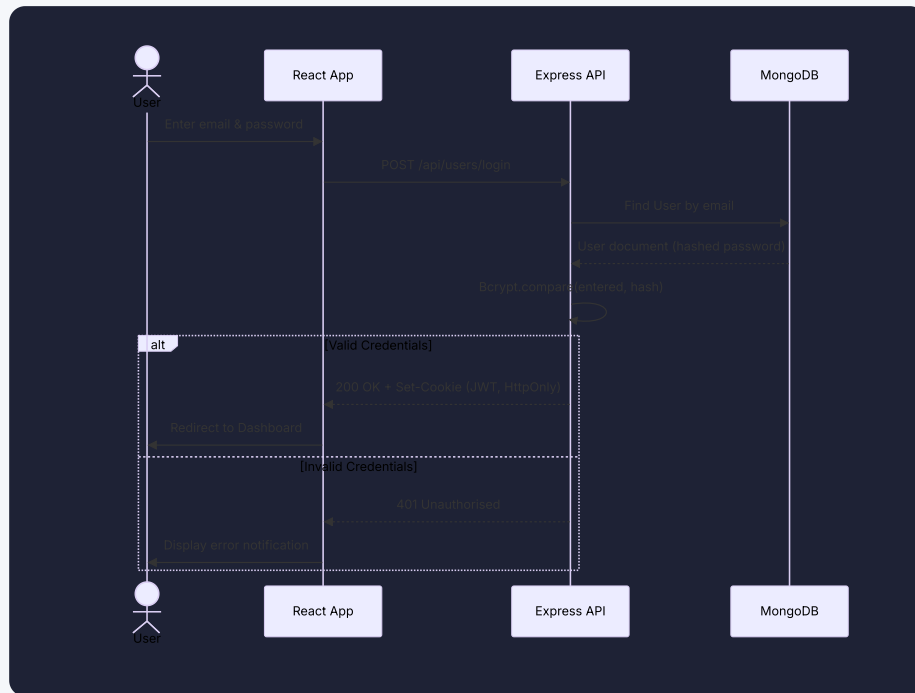


FIGURE 11.4 — AUTHENTICATION SEQUENCE DIAGRAM

6. Setup & Installation

6.1 Environment Variables (`server/.env`)

```
PORT=8000
MONGO_URI=mongodb+srv://<user>:<password>@<cluster>.mongodb.r
JWT_SECRET=your_secret_key_here
STRIPE_SECRET_KEY=sk_test_your_stripe_key
CLOUDINARY_CLOUD_NAME=your_cloud_name
CLOUDINARY_API_KEY=your_api_key
CLOUDINARY_API_SECRET=your_api_secret
```

6.2 Installation Commands

```
# Install server dependencies & seed DB
cd server && npm install && npm run seed && npm run dev

# Install client dependencies
cd ../client && npm install && npm run dev
```

7. Complete API Reference

Method	Endpoint	Auth	Description
POST	/api/users/register	No	Register a new user
POST	/api/users/login	No	Authenticate and issue JWT
POST	/api/users/logout	Yes	Clear JWT cookie
GET	/api/users/profile	User	Get user profile
PUT	/api/users/profile	User	Update user profile
GET	/api/products	No	Get products (supports filters)
GET	/api/products/:id	No	Get single product
POST	/api/products	Admin	Create product
PUT	/api/products/:id	Admin	Update product
DELETE	/api/products/:id	Admin	Delete product
POST	/api/products/:id/reviews	User	Submit a review
POST	/api/orders	User	Create new order
GET	/api/orders/myorders	User	Get user's order history
GET	/api/orders/:id	User/Admin	Get order by ID
PUT	/api/orders/:id/pay	User	Mark order as paid

Method	Endpoint	Auth	Description
PUT	/api/orders/:id/deliver	Admin	Mark order as delivered
GET	/api/orders	Admin	Get all orders

8. Known Issues & Future Enhancements

Type	ID	Description	Mitigation / Plan
Known Issue	KI-01	Cart state may desync on concurrent multi-device sessions	Server-side <code>updatedAt</code> timestamp comparison on cart save
Known Issue	KI-02	Render free-tier cold-start causes 30–60s initial delay	Upgrade to paid tier for production
Enhancement	FE-01	Email notifications for order confirmation and shipping updates	Nodemailer / SendGrid — Future Release
Enhancement	FE-02	AI-driven personalised product recommendations	Collaborative filtering — Future Roadmap
Enhancement	FE-03	Multi-vendor marketplace with revenue-share model	Significant architectural change — Future Roadmap

9. Live Deployment

Resource	URL / Credentials
Live Application	https://e-commerce-application-neon-five.vercel.app/
API Base URL	https://shopez-api-c30e.onrender.com
Admin Account	admin@shopez.com / adminpassword123
Customer Account	test@email.com / password123